Introduction to Cybersecurity

Cybersecurity Lab-2

Module 2 Network Discovery and Scanning with Nmap

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Module 2 Network Discovery and Scanning with Nmap

Part 1 Verify Installation

The command nmap --version was executed successfully, confirming that Nmap is working correctly.

```
Windows PowerShell

PS C:\Users\Brecheyn> nmap --version
Nmap version 7.98 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.17 nmap-libssh2-1.11.1 nmap-lib
z-1.3.1 nmap-libpcre2-10.45 Npcap-1.86 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select
PS C:\Users\Brecheyn> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 5:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::88e5:7da2:9e1f:3dbc%13
   IPv4 Address. . . . . . . . . . . : 192.168.100.148
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.100.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
PS C:\Users\Brecheyn> nmap -sn 192.168.100.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:22 +0000
Nmap scan report for 192.168.100.1
```

Part 2 Discover Active Hosts

The command nmap -sn 192.168.100.0/24 was used to scan the network

```
Windows PowerShell
   Connection-specific DNS Suffix  . :
PS C:\Users\Brecheyn> nmap -sn 192.168.100.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:22 +0000
Nmap scan report for 192.168.100.1
Host is up (0.047s latency).
MAC Address: 48:4C:29:38:68:83 (Huawei Technologies)
Nmap scan report for 192.168.100.5
Host is up (0.048s latency).
MAC Address: F8:E4:E3:40:11:BD (Intel Corporate)
Nmap scan report for 192.168.100.7
Host is up (0.079s latency).
MAC Address: 32:E8:7C:2F:D5:25 (Unknown)
Nmap scan report for 192.168.100.14
Host is up (0.093s latency).
MAC Address: FE:B9:3D:AC:16:88 (Unknown)
Nmap scan report for 192.168.100.33
Host is up (0.12s latency).
MAC Address: A2:28:76:C6:24:C5 (Unknown)
Nmap scan report for 192.168.100.78
Host is up (0.11s latency).
MAC Address: 32:03:17:30:8A:34 (Unknown)
Nmap scan report for 192.168.100.79
Host is up (0.039s latency).
MAC Address: 72:2A:E2:61:5D:B9 (Unknown)
Nmap scan report for 192.168.100.85
Host is up (0.25s latency).
MAC Address: 04:ED:33:10:E0:F0 (Intel Corporate)
Nmap scan report for 192.168.100.135
Host is up (0.10s latency).
MAC Address: 0E:B4:95:07:DE:DC (Unknown)
Nmap scan report for 192.168.100.138
Host is up (0.12s latency).
MAC Address: 90:78:41:C4:CE:8A (Intel Corporate)
Nmap scan report for 192.168.100.146
Host is up (0.048s latency).
MAC Address: DC:45:46:19:A5:28 (Intel Corporate)
Nmap scan report for 192.168.100.154
Host is up (0.11s latency).
MAC Address: 3E:CA:27:00:B8:43 (Unknown)
Nmap scan report for 192.168.100.155
Host is up (0.11s latency).
MAC Address: F8:71:0C:15:80:6C (Xiaomi Communications)
Nmap scan report for 192.168.100.156
```

. The results showed that 28 hosts are active.

Examples of active hosts include

1– 192.168.100.1

2- 192.168.100.155.

Part 3 Scan Open Ports

```
Windows PowerShell
PS C:\Users\Brecheyn> nmap 192.168.100.155
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:25 +0000
Nmap scan report for 192.168.100.155
Host is up (0.037s latency).
All 1000 scanned ports on 192.168.100.155 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: F8:71:0C:15:80:6C (Xiaomi Communications)

Nmap done: 1 IP address (1 host up) scanned in 10.83 seconds
PS C:\Users\Brecheyn> nmap 192.168.100.229
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:26 +0000
Nmap scan report for 192.168.100.229
Host is up (0.054s latency).
All 1000 scanned ports on 192.168.100.229 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: E4:70:B8:99:D2:2F (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 64.45 seconds
PS C:\Users\Brecheyn> nmap 192.168.100.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:29 +0000
Nmap scan report for 192.168.100.1
Host is up (0.012s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE    SERVICE
21/tcp   filtered ftp
22/tcp   filtered ssh
23/tcp   filtered telnet
53/tcp   open     domain
80/tcp   open     http
8022/tcp filtered oa-system
MAC Address: 48:4C:29:38:68:83 (Huawei Technologies)

Nmap done: 1 IP address (1 host up) scanned in 7.56 seconds
PS C:\Users\Brecheyn> nmap  -sV 192.168.100.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:32 +0000
Nmap scan report for 192.168.100.1
Host is up (0.039s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE    SERVICE   VERSION
21/tcp   filtered ftp
22/tcp   filtered ssh
23/tcp   filtered telnet
53/tcp   open     domain    (generic dns response: NOTIMP)
```

```
23/tcp   filtered telnet
53/tcp   open     domain
80/tcp   open     http
8022/tcp filtered oa-system
MAC Address: 48:4C:29:38:68:83 (Huawei Technologies)

Nmap done: 1 IP address (1 host up) scanned in 7.56 seconds
```

Port scanning was performed on selected hosts. Some hosts such as 192.168.100.155 and 192.168.100.229 did not show open ports because all ports were closed or filtered.

A successful scan was obtained on 192.168.100.1.

The results showed that port

1 . 53 is open and used for DNS,

2. **80 is open and used for HTTP**.

3. Ports 21, 22, 23 and 8022 are filtered, indicating restricted access.

Part 4 Service Detection

```
Windows PowerShell
PS C:\Users\Brecheyn> nmap  -sV 192.168.100.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:32 +0000
Nmap scan report for 192.168.100.1
Host is up (0.039s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE    SERVICE   VERSION
21/tcp   filtered ftp
22/tcp   filtered ssh
23/tcp   filtered telnet
53/tcp   open     domain    (generic dns response: NOTIMP)
80/tcp   open     ssl/http
8022/tcp filtered oa-system
1 service unrecognized despite returning data. If you know the service/versi
on, please submit the following fingerprint at https://nmap.org/cgi-bin/subm
it.cgi?new-service :
SF-Port53-TCP:V=7.98%I=7%D=3/23%Time=69C16B1E%P=i686-pc-windows-windows%r(
SF:DNSStatusRequestTCP,E,"\0\x0c\0\0\x90\x04\0\0\0\0\0\0\0\0");
MAC Address: 48:4C:29:38:68:83 (Huawei Technologies)

Service detection performed. Please report any incorrect results at https://
nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 48.49 seconds
PS C:\Users\Brecheyn> nmap -o 192.168.100.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:33 +0000
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.04 seconds
PS C:\Users\Brecheyn> nmap -0 192.168.100.1
C:\Program Files (x86)\Nmap\nmap.exe: illegal option -- 0
See the output of nmap -h for a summary of options.
PS C:\Users\Brecheyn> nmap -O 192.168.100.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:35 +0000
Nmap scan report for 192.168.100.1
Host is up (0.033s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE    SERVICE
21/tcp   filtered ftp
22/tcp   filtered ssh
23/tcp   filtered telnet
53/tcp   open     domain
80/tcp   open     http
8022/tcp filtered oa-system
MAC Address: 48:4C:29:38:68:83 (Huawei Technologies)
Device type: general purpose
```

The command nmap -sV 192.168.100.1 was used to detect services.

The results confirmed that port 53 runs a DNS service and port 80 runs an HTTP service secured with SSL. The version details were partially limited due to filtering.

Part 5 OS Detection

```
Windows PowerShell
PS C:\Users\Brecheyn> nmap -O 192.168.100.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:35 +0000
Nmap scan report for 192.168.100.1
Host is up (0.033s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE    SERVICE
21/tcp   filtered ftp
22/tcp   filtered ssh
23/tcp   filtered telnet
53/tcp   open     domain
80/tcp   open     http
8022/tcp filtered oa-system
MAC Address: 48:4C:29:38:68:83 (Huawei Technologies)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.10 - 4.11
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.
org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.29 seconds
```

The command nmap -O 192.168.100.1 was used to detect the operating system.

The results indicated that the target device is running a Linux based operating system, specifically between versions 3.10 and 4.11.

This is consistent with the device being a Huawei router. The network distance to the device is one hop, confirming that it is on the local network.

Part 6 Aggressive Scan

```
Windows PowerShell
PS C:\Users\Brecheyn> nmap -A 192.168.100.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 16:36 +0000
Nmap scan report for 192.168.100.1
Host is up (0.014s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE    SERVICE   VERSION
21/tcp   filtered ftp
22/tcp   filtered ssh
23/tcp   filtered telnet
53/tcp   open     domain    (generic dns response: NOTIMP)
80/tcp   open     ssl/http
|_ssl-date: 2026-03-23T16:12:22+00:00; -25m03s from scanner time.
| ssl-cert: Subject: commonName=ont.huawei.com/organizationName=Huawei/count
ryName=CN
| Not valid before: 2020-08-25T06:59:49
|_Not valid after:  2030-08-23T06:59:49
|_http-title: Site doesn't have a title (text/html; charset=UTF-8).
8022/tcp filtered oa-system
1 service unrecognized despite returning data. If you know the service/versi
on, please submit the following fingerprint at https://nmap.org/cgi-bin/subm
it.cgi?new-service :
SF-Port53-TCP:V=7.98%I=7%D=3/23%Time=69C16C18%P=i686-pc-windows-windows%r(
SF:DNSStatusRequestTCP,E,"\0\x0c\0\0\x90\x04\0\0\0\0\0\0\0\0");
MAC Address: 48:4C:29:38:68:83 (Huawei Technologies)
Device type: general purpose
Running: Linux 3.X|4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4 cpe:/o:linux
:linux_kernel:5
OS details: Linux 3.10 - 4.11, Linux 5.10 - 5.13
Network Distance: 1 hop

Host script results:
|_clock-skew: -25m03s

TRACEROUTE
HOP RTT      ADDRESS
1   14.50 ms 192.168.100.1

OS and Service detection performed. Please report any incorrect results at h
ttps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 68.96 seconds
PS C:\Users\Brecheyn>
```

The command nmap -A 192.168.100.1 was executed to perform an advanced scan.

This scan combines service detection, OS detection and script scanning.

<u>The results confirmed the presence of DNS and HTTP services and revealed SSL certificate information associated with Huawei.</u>

Analysis Questions

1. A total of 28 hosts were detected on the network.

2. The most common and important ports identified are port 53 for DNS and port 80 for HTTP. These ports are essential for communication and web services.

3. Open ports are a security risk because they expose services that attackers can exploit.

4. Attackers can gather information such as active hosts, open and filtered ports, running services and the operating system of network devices.